How Lexodle scores a guess, in plain terms

The Lexodle project

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The maths here is mostly AI-generated and has not been peer reviewed.

This is the same material as the paper `main.pdf`, with the notation taken out. You need fractions, percentages and averages. For the logarithm, one calculator button (log base 2) is enough, and every number in the examples can be checked with it.

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1 The game, and what is being measured

A word is hidden. You guess a word of the same length, and each letter comes back green (right place), yellow (in the word, wrong place) or grey (not in the word). You keep guessing until you find the word or run out of tries.

Every guess is a question and the colours are the answer. Before a guess some words are still possible. After it, only the words that would have given *exactly those colours* are still possible. A good guess leaves few words, and the answer tells you a lot.

Lexodle measures that in **bits**. A bit is what it takes to cut the possible words in half. Eight words left to one is three bits; 1,024 to one is ten. Finding the word means gaining all the bits there are to find: for N equally likely words, $\log_2 N$ of them.

2 Bits

A guess splits the possible words into groups, one for each set of colours it could get. If you land in a group that held a share q of the chances, you **gain** $-\log_2 q$ bits. A group holding half the chances gains one bit; one holding a quarter gains two; one holding a thousandth gains about ten.

Example. Three words, equally likely, and a guess that tells apart one of them from the other two. Groups: one word ($q = 1/3$, gain 1.58 bits) and two words ($q = 2/3$, gain 0.58 bits).

The **expected gain**, written H, is each outcome's gain times its chance, added up: $\frac{1}{3} \times 1.58 + \frac{2}{3} \times 0.58 = 0.92$ bits. It is what the guess is worth before you know the colours.

3 Skill

Every word the game accepts could have been your guess. The best of them has the largest expected gain, H^*. Your **skill** is your guess's H as a share of H^*, as a percentage.

In the example, suppose the best guess could gain 1.58 bits (every word alone). Then the guess above has skill $0.92/1.58 \times 100 = 58$. A guess that tells you nothing (one group) has skill 0.

Skill does not depend on the hidden word. It depends only on what was known and on the guess.

4 Luck

The colours you got gained some number of bits. Luck says where that gain falls among the gains the guess could have given: the chance of a smaller gain, minus the chance of a larger one, as a percentage.

In the example, landing in the single word (gain 1.58, the biggest, a 1/3 chance) has luck $2/3 - 0 = +67$. Landing in the pair (gain 0.58, a 2/3 chance) has luck $0 - 1/3 = -33$. Across

all the ways the answer could be, luck averages exactly 0: it is neither a reward nor a penalty, only how the colours fell.

Luck is also shown in bits: your gain less the expected gain. Here +0.67 bits for the single word, −0.33 for the pair. Added up over a game, this is the part of your bits that was luck.

5 Risk

Some guesses gain about the same whatever the colours. Others usually gain a little and now and then a lot: a gamble. Risk is the chance of gaining less than expected, minus the chance of gaining more, as a percentage of the riskiest guess there was (never below 0).

In the example the chance of gaining less than 0.92 is 2/3 and of gaining more 1/3, so the raw risk is +33 points. A guess with equal groups has none. Risk belongs to the guess, not to how it turned out, so it is the same however lucky you were.

6 Your whole game

The bits you gain guess by guess add up. When you find the word you have gained all the bits there were to find. For a game: bits gained = bits expected (your skill) + luck. A game's picture is a bar for each guess racing towards the bits needed.

7 The last try

On the last try nothing is left to learn: only a word that fits can win. Skill is the chance your guess wins, against the likeliest word's. Luck is the outcome less that chance: if three words fit and you guess one, a win is +67 and a miss is −33.

8 Several words at once

In a quartet one guess serves four boards. Each board is scored on its own, and the guess's skill and luck are the averages over the boards that are still open, each counted by the bits it still has to find.

9 The other modes

The scores are the same in the other single-word modes. In **lockdown** a guess must keep the greens where they are, so skill is measured against the best word you were *allowed*, not the best in the whole list. In **blind** you only get two counts (letters right and in place, letters right elsewhere), so a group is the words that give the same two counts.

10 Playstyles

Your style is read from your last 100 classic games (a classic daily counts). Early on (64 or more words left) each guess counts as the first that fits: a **gambler** (risk 75 or more), a **maximiser** (in the top 10% of guesses by expected gain), an **eliminator** (four or more letters never tried). At the end (16 or fewer words, or three greens known) each guess counts as a **closer** (it fits every clue, so it could win) or a **prober** (it does not, ranks in the top 10%, and three or more words are left).

Your **habit** in a phase is what at least 40% of its guesses were, and the most common if several are. Your style is your early habit and your end habit:

	prober at the end	closer at the end
maximiser early	analyst	maverick
eliminator early	detective	sweeper
gambler early	wildcard	gambler

You are rated after ten games with ten guesses in each phase. These thresholds come from simulated players, not from real ones, and may change.